

Youssef Hanna

Mechanical Engineering | Aerospace • Mechatronics • Embedded Systems

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Portfolio: [youssefhanna.vercel.app](#) | GitHub: [github.com/TheLeg336](#)

EDUCATION

California State Polytechnic University, Pomona

Expected May 2028

Bachelor of Science in Mechanical Engineering | GPA: 3.74

Relevant Coursework: Statics, Dynamics, Mechanics of Materials, Thermodynamics, Engineering Graphics (SolidWorks)

ENGINEERING PROJECTS

Precision Tennis Ball Launcher — Project Lead

2024

Intro to Engineering & Design | 1st Place — Class Competition

- Led a ~15-person team through design, fabrication, testing, and cost planning for a competitive mechanical launcher.
- Recovered from a major design failure one week before competition by coordinating a redesign around altered components, producing a more compact final configuration.
- Achieved ~300 ft accurate range vs. a ~100 ft requirement (3× target) while maintaining an estimated 30–40% project profit margin and winning the class competition.

DualSense PC Interface — Designer / Builder

2026

Raspberry Pi Pico 2 W (RP2350) | Embedded Systems & Electronics

- Built a Raspberry Pi Pico 2 W-based hardware interface that allows a DualSense controller to communicate with a PC as a wired connection while preserving controller inputs and advanced feedback features.
- Integrated and debugged haptics, adaptive triggers, microphone/audio functionality, device communication, and soldered electronics.

UniRate — Solo Developer

2026

JavaScript, HTML/CSS, Python | Published Chrome Extension

- Designed, built, and published a Chrome extension that integrates Rate My Professors information directly into university registration and class-search portals.
- Developed professor matching, ratings, difficulty and would-take-again information, course-specific review filtering, rating thresholds, and multi-school support; used AI-assisted development for code review, debugging, and expansion.

Ezer — Local AI Computer Assistant

2026–Present

Tauri, Rust, SolidJS, SQLite | Independent Project (In Development)

- Conceived and designed a local Windows computer-use assistant, defining system requirements, workflows, safety/confirmation behavior, edge cases, and testing while directing AI-assisted implementation.
- Exploring lightweight local models and engineering-software workflows, including future SolidWorks computer-use capability.

TECHNICAL SKILLS

CAD & Mechanical Design: SolidWorks, Onshape, Blender

Embedded & Fabrication: Raspberry Pi Pico 2 W, Soldering, 3D Printing (FDM), Laser Cutting, Electronics Integration, Hand & Power Tools

Programming & Software: Python, C++, JavaScript, HTML/CSS

Tools & Languages: Excel | English & Arabic (fluent)

EXPERIENCE & SERVICE

Kero Security — Security Guard

Mar 2026 – Aug 2026

- Monitored assigned sites, performed routine patrols, communicated with staff and visitors, and helped maintain a safe environment while working independently.

Volunteer Mathematics Tutor — Church & Community

- Tutored mathematics for students and community members, breaking complex concepts into clear step-by-step explanations.

HONORS

Dean's List | Phi Theta Kappa Honor Society